AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 6: Second-life grading, application matching and value

Mon 7 Dec - Fri 11 Dec 2026 · Sprint 6 · Overall progress 66%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 11 Dec 2026

On track **This week in one line:** Uncertainty-aware grader agrees with lab-truth grades in 83.2% of 101 decisions (89.5% when confident) and flags 65% of its own mistakes for a lab test.

1. Goal for the week

Decide what each retired battery should do next, with honest confidence, and quantify the value of re-use.

2. Day-by-day work

Day	Work done
Mon 7 Dec	Application catalogue: A (EV re-use), B (grid/home storage), C (backup power), D (recycle) with SOH, resistance and cycle limits.
Tue 8 Dec	Rule engine with reasons; confidence check by re-grading at the lower 95% bound of SOH.
Wed 9 Dec	Service years, CO2 avoided and value, scaled to a 60 kWh pack.
Thu 10 Dec	Evaluated at 15 retirement levels per cell against lab-truth grades; confusion matrix.
Fri 11 Dec	Figure, 12 new tests (33 total), this report.

3. Completed deliverables

- **batterylife/lifecycle/**: grading.py, sustainability.py, second_life.py.
- **results/second_life_assessments.csv** and summary.
- tests/test_lifecycle.py.

4. Results

Metric	Result
Grade agreement with lab truth	83.2% (NASA 87.8%, CALCE 80.0%)
Agreement when confidence is high	89.5%
Mistakes already flagged low-confidence	64.7%
Conservative / optimistic errors	7 / 10
Second-life length error (CALCE)	33.6 cycles
Per 60 kWh pack at 80% retirement	about 3.6 t CO2e avoided (upper bound), USD 2,478 re-use value

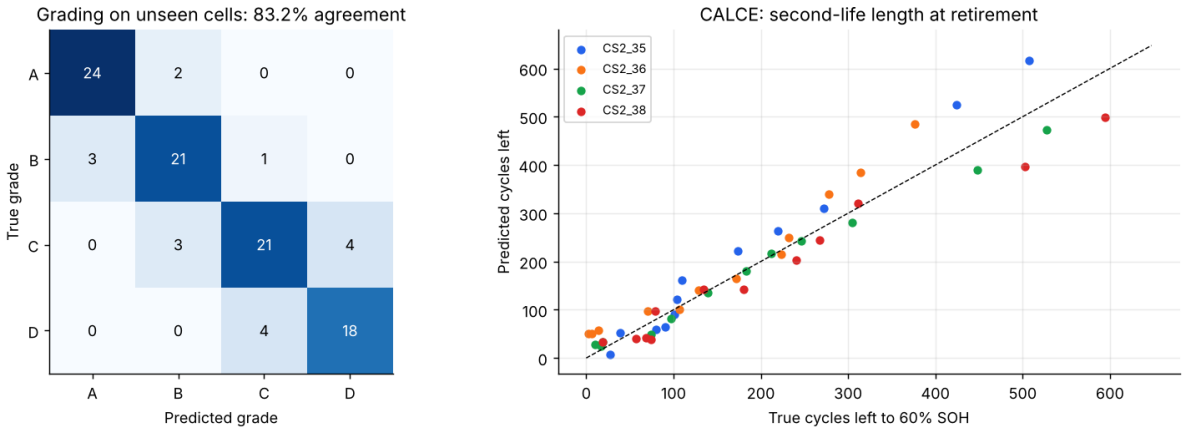


Figure 6. Left: AI grade vs lab grade on unseen cells. Right: predicted vs true second-life length.

5. Problems faced and how they were solved

Problem	Resolution
Most wrong grades sit right at a grade boundary.	The low-confidence flag catches them and asks for a lab test.
Only one NASA cell reaches 60% SOH.	Cycle criterion applied only where ground truth exists (CALCE).
Lab ageing is harsh, so service years look short.	Reported as a relative ranking, not a forecast.

6. Plan for next week

- Build the signed, hash-chained Digital Battery Passport.
- Test it against tampering attacks.

Faculty remarks

Faculty signature: _____ Student signature: _____